CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 15: Similarity (With Applications to Maps and Models)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Similar triangles

1. $\triangle ABC \sim \triangle EFG$ with $\angle ABC = \angle EFG = 60°$, $AB = 15$ cm, $BC = 3$ cm and $EF = 75$ cm. Find FG.
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm ICSE 2025, 1 mark

SOLUTION

Corresponding sides are proportional: $\frac{AB}{EF} = \frac{BC}{FG}$.

$\frac{15}{75} = \frac{3}{FG}$, so $FG = 15$ cm.

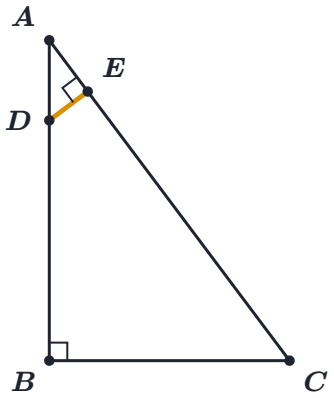
Answer: (a)

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2 Maps and models

2. In $\triangle ABC$, $\angle ABC = 90^\circ$, $AB = 20$ cm and $AC = 25$ cm. D is a point on AB and E is a point on AC with $DE \perp AC$ and $DE = 3$ cm. (a) Prove that $\triangle ABC \sim \triangle AED$. (b) Find BC, AD and AE. (c) $BCED$ is a plot of land drawn on a map, and its actual area is 576 m². Find the scale factor of the map.

ICSE 2025, 4 marks



SOLUTION

(a) In $\triangle ABC$ and $\triangle AED$: $\angle A$ is common and $\angle ABC = \angle AED = 90^\circ$. So $\triangle ABC \sim \triangle AED$ (AA).

(b) $BC = \sqrt{25^2 - 20^2} = 15$ cm. From the similar triangles, $\frac{AB}{AE} = \frac{BC}{ED} = \frac{AC}{AD} = \frac{15}{3} = 5$.

$AE = \frac{20}{5} = 4$ cm and $AD = \frac{25}{5} = 5$ cm.

(c) Area of $BCED$ on the map $=$ area of $\triangle ABC -$ area of $\triangle AED = \frac{1}{2} \times 20 \times 15 - \frac{1}{2} \times 4 \times 3 = 150 - 6 = 144$ cm².

Actual area $= 576$ m² $= 5760000$ cm². $k^2 = \frac{144}{5760000} = \frac{1}{40000}$, so $k = \frac{1}{200}$.

Answer: (b) $BC = 15$ cm, $AD = 5$ cm, $AE = 4$ cm (c) scale factor $1 : 200$

3. A model of a high-rise building is made to a scale of $1 : 50$. (a) The model is 0.8 m high; find the height of the building. (b) A flat in the building has a floor area of 20 m²; find its floor area in the model.

ICSE 2020, 3 marks

SOLUTION

(a) Height of building $= 0.8 \times 50 = 40$ m

(b) Areas are in the ratio $1^2 : 50^2 = 1 : 2500$. Area in the model $= \frac{20}{2500} = 0.008$ m² $= 80$ cm².

Answer: (a) 40 m (b) 0.008 m², that is 80 cm²

4. A model of a building is made with scale factor $1 : 30$. (a) The model is 80 cm high; find the height of the building in metres. (b) A tank on top of the building has a volume of 27 m³; find the volume of the tank on the model. ICSE 2019, 3 marks

SOLUTION

(a) Height $= 80 \times 30 = 2400$ cm $= 24$ m

(b) Volumes are in the ratio $1^3 : 30^3 = 1 : 27000$. Volume on the model $= \dfrac{27}{27000} = 0.001$ m³ $= 1000$ cm³.

Answer: (a) 24 m (b) 0.001 m³, that is 1000 cm³

5. On a map drawn to a scale of $1 : 50{,}000$, a rectangular plot $ABCD$ has $AB = 6$ cm and $BC = 8$ cm. Find (a) the actual length of the diagonal AC in km (b) the actual area of the plot in km². ICSE 2018, 3 marks

SOLUTION

1 cm on the map is 50,000 cm $= 0.5$ km on the ground.

(a) $AC = \sqrt{6^2 + 8^2} = 10$ cm on the map, which is $10 \times 0.5 = 5$ km.

(b) The actual sides are 3 km and 4 km, so the area is $3 \times 4 = 12$ km².

Answer: (a) 5 km (b) 12 km²

3 Conditions for similarity

6. $\triangle ABC \sim \triangle PQR$. AD bisects $\angle BAC$ and meets BC at D; PS bisects $\angle QPR$ and meets QR at S. Which statement is true? (a) $\triangle ABC \sim \triangle PQS$ (b) $\triangle ABD \sim \triangle PQS$ (c) $\triangle ABD \sim \triangle PSR$ (d) $\triangle ABC \sim \triangle PSR$ ICSE 2024, 1 mark

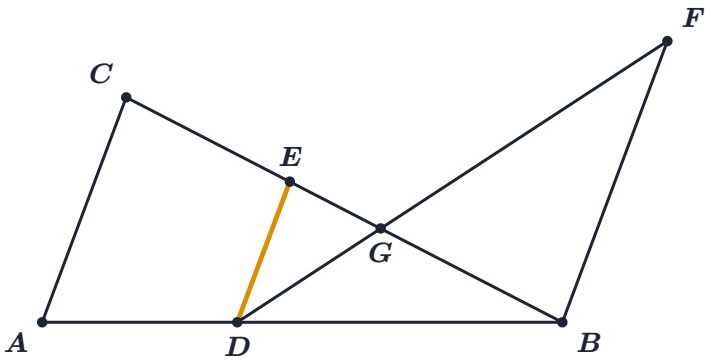
SOLUTION

$\angle B = \angle Q$ and $\angle A = \angle P$ (similar triangles), so $\angle BAD = \dfrac{1}{2}\angle A = \dfrac{1}{2}\angle P = \angle QPS$.

Two angles of $\triangle ABD$ equal two angles of $\triangle PQS$.

Answer: (b)

7. AC, DE and BF are parallel, with D on AB and E on CB. DF meets CB at G. Given $AC = 24$ cm, $EG = 8$ cm, $GB = 16$ cm and $BF = 30$ cm, (a) prove that $\triangle GED \sim \triangle GBF$ (b) find DE (c) find $DB : AB$. ICSE 2023, 4 marks



SOLUTION

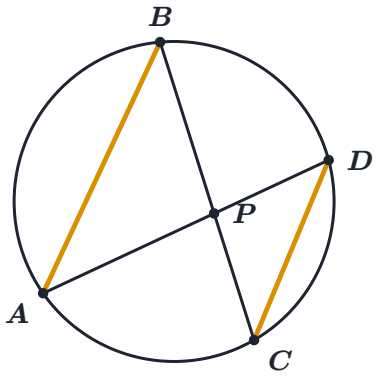
(a) $\angle EGD = \angle BGF$ (vertically opposite) and $\angle GED = \angle GBF$ (alternate angles, $DE \parallel BF$). So $\triangle GED \sim \triangle GBF$ (AA).

(b) $\frac{DE}{BF} = \frac{EG}{GB}$, so $DE = 30 \times \frac{8}{16} = 15$ cm.

(c) In $\triangle ABC$, $DE \parallel AC$, so $\triangle BDE \sim \triangle BAC$ and $\frac{DB}{AB} = \frac{DE}{AC} = \frac{15}{24} = \frac{5}{8}$.

Answer: (b) $DE = 15$ cm (c) $DB : AB = 5 : 8$

8. Chords AD and BC of a circle intersect at P. $AB = 9$ cm, $PA = 7.5$ cm and $PC = 5$ cm. (a) Prove that $\triangle PAB \sim \triangle PCD$. (b) Find CD. (c) Find area of $\triangle PAB$: area of $\triangle PCD$. ICSE 2020, 3 marks



SOLUTION

(a) $\angle APB = \angle CPD$ (vertically opposite) and $\angle PAB = \angle PCD$ (angles in the same segment, standing on the arc BD). So $\triangle PAB \sim \triangle PCD$ (AA).

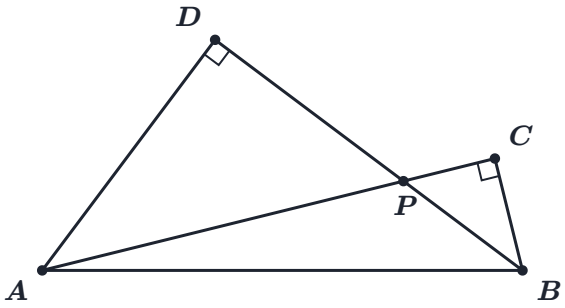
(b) $\frac{AB}{CD} = \frac{PA}{PC}$, so $CD = \frac{9 \times 5}{7.5} = 6$ cm.

(c) Ratio of areas $= \left(\frac{PA}{PC}\right)^2 = \left(\frac{7.5}{5}\right)^2 = \frac{9}{4}$

Answer: (b) $CD = 6$ cm (c) $9 : 4$

4 Areas of similar triangles

9. $\triangle ADB$ and $\triangle ACB$ are right-angled at D and C, on the same side of AB, and AC meets BD at P. $AB = 10$ cm, $AD = 6$ cm, $BC = 2.4$ cm and $DP = 4.5$ cm. (a) Prove that $\triangle APD \sim \triangle BPC$. (b) Find BD and PB. (c) Hence find PA. (d) Find area of $\triangle APD$: area of $\triangle BPC$. ICSE 2024, 4 marks



SOLUTION

(a) $\angle ADP = \angle BCP = 90^\circ$ and $\angle APD = \angle BPC$ (vertically opposite). So $\triangle APD \sim \triangle BPC$ (AA).

(b) In $\triangle ADB$: $BD = \sqrt{10^2 - 6^2} = 8$ cm. $PB = BD - DP = 8 - 4.5 = 3.5$ cm.

(c) From the similar triangles, $\frac{PA}{PB} = \frac{AD}{BC} = \frac{6}{2.4} = 2.5$, so $PA = 2.5 \times 3.5 = 8.75$ cm.

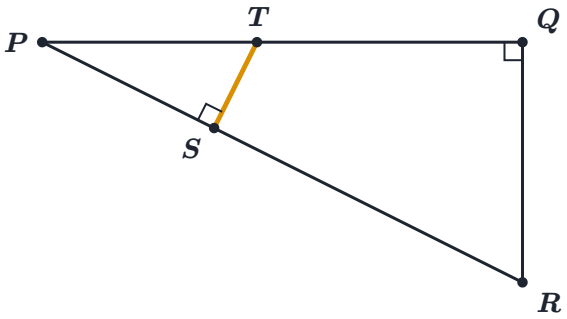
(d) Ratio of areas $= \left(\frac{AD}{BC}\right)^2 = (2.5)^2 = \frac{25}{4}$

Answer: (b) $BD = 8$ **cm,** $PB = 3.5$ **cm (c)** $PA = 8.75$ **cm (d)** $25 : 4$

Note: the measurements in this question do not fit together exactly (the Pythagoras theorem in $\triangle ADP$ would give $PA = 7.5$ cm). The word "hence" tells you to use the similar triangles, as done here.

10. In $\triangle PQR$, $\angle PQR = 90^\circ$. T is a point on PQ and S is a point on PR with $\angle PST = 90^\circ$. $PQ = 5$ cm and $PS = 2$ cm. (a) Prove that $\triangle PQR \sim \triangle PST$. (b) Find area of $\triangle PQR$: area of quadrilateral $SRQT$.

ICSE 2019, 3 marks



SOLUTION

(a) $\angle P$ is common and $\angle PQR = \angle PST = 90^\circ$. So $\triangle PQR \sim \triangle PST$ (AA).

(b) $$\frac{\text{area of } \triangle PQR}{\text{area of } \triangle PST} = \left(\frac{PQ}{PS}\right)^2 = \frac{25}{4}$$

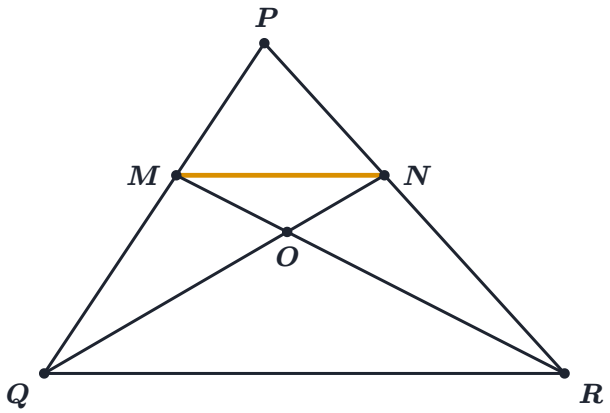
If the area of $\triangle PQR$ is 25 parts, the area of $\triangle PST$ is 4 parts and the quadrilateral $SRQT$ is $25 - 4 = 21$ parts.

Answer: $25 : 21$

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11. In $\triangle PQR$, M lies on PQ and N lies on PR with $MN \parallel QR$ and $\frac{PM}{MQ} = \frac{2}{3}$. MR and NQ meet at O. (a) Find $\frac{MN}{QR}$. (b) Prove that $\triangle OMN \sim \triangle ORQ$. (c) Find area of $\triangle OMN$: area of $\triangle ORQ$.

ICSE 2018, 3 marks



SOLUTION

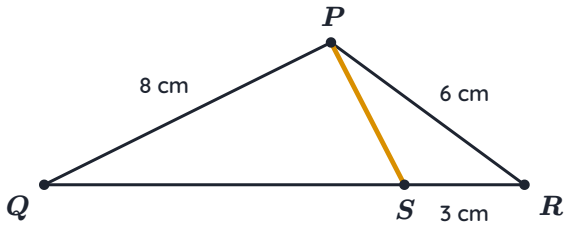
(a) $\triangle PMN \sim \triangle PQR$ (AA, because $MN \parallel QR$), so $\frac{MN}{QR} = \frac{PM}{PQ} = \frac{2}{2+3} = \frac{2}{5}$.

(b) $\angle MON = \angle ROQ$ (vertically opposite) and $\angle OMN = \angle ORQ$ (alternate angles). So $\triangle OMN \sim \triangle ORQ$ (AA).

(c) Ratio of areas $= \left(\frac{MN}{QR}\right)^2 = \frac{4}{25}$

Answer: (a) $\frac{2}{5}$ (c) $4 : 25$

12. S is a point on the side QR of $\triangle PQR$ with $\angle PSR = \angle QPR$. $QP = 8$ cm, $PR = 6$ cm and $SR = 3$ cm. (a) Prove that $\triangle PQR \sim \triangle SPR$. (b) Find QR and PS. (c) Find $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle SPR}$. ICSE 2017, 3 marks



SOLUTION

(a) $\angle R$ is common and $\angle QPR = \angle PSR$ (given). So $\triangle PQR \sim \triangle SPR$ (AA).

(b) $\frac{PQ}{SP} = \frac{QR}{PR} = \frac{PR}{SR} = \frac{6}{3} = 2$. So $QR = 2 \times 6 = 12$ cm and $PS = \frac{8}{2} = 4$ cm.

(c) Ratio of areas $= 2^2 = 4$

Answer: (b) $QR = 12$ cm, $PS = 4$ cm (c) 4

5 Basic proportionality theorem

13. In $\triangle PAB$, C is a point on PA and D is a point on PB with $\angle BAP = \angle DCP = 70^\circ$. If $PC = 6$ cm and $CA = 4$ cm, find $PD : DB$. (a) $5 : 3$ (b) $3 : 5$ (c) $3 : 2$ (d) $2 : 3$ ICSE 2023, 1 mark

SOLUTION

The equal angles are corresponding angles, so $CD \parallel AB$.

By the basic proportionality theorem, $\frac{PD}{DB} = \frac{PC}{CA} = \frac{6}{4} = \frac{3}{2}$.

Answer: (c)

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